Sentira-Torch AI: Performance Report

1. Student Information

- **Name:** Mahmoud Khamis Belal Abdelrady
- **University:** Badr University in Assiut (BUA)
- **Faculty:** Faculty of Artificial Intelligence
- **Department:** Data Science - 3rd Year

- **Framework:** PyTorch & Gradio

2. Methodology & Reproducibility

The system uses TF-IDF vectorization (5,000 features) with custom lemmatization. To ensure stability, a global Seed (42) was implemented across PyTorch, NumPy, and Python. Deployment was handled via Gradio for real-time inference.

3. Comparative Results

Metric	Experiment 1 (32 Neurons)	Experiment 2 (64 Neurons)
Learning Rate	0.001	0.0007
Batch Norm	No	Yes
Peak Accuracy	87.42%	87.73%
Final Test Acc	86.96%	87.61%
Final Test Loss	0.3288	0.3019

4. Conclusion

Experiment 2 is the superior architecture. By utilizing 64 neurons and Batch Normalization with a stabilized seed, the model achieved better generalization and a lower error rate (Loss: 0.3019), making it highly reliable for sentiment prediction.